

SHIVAM PATIL

[+91 7038573273](tel:+917038573273) | shivampatilinfo@gmail.com | linkedin.com/in/shivam--patil/ | shivamsfolio.com

PROFESSIONAL SUMMARY:

Software Engineer with 1.5+ years of experience designing high-throughput distributed systems and delivering production-grade applications.

TECHNICAL SKILLS:

Languages: C++, Python, Go, Rust, JavaScript, TypeScript

Frontend: React.js, Next.js

Backend: FastAPI, REST APIs, Microservices, Distributed Systems

Cloud & DevOps: Azure Kubernetes Service (AKS), AWS, Docker, Kubernetes, Prometheus, Grafana

AI/ML & Generative AI: AI Agents, Prompt Engineering, RAG (Retrieval-Augmented Generation), Vector Databases

Core CS: System Programming, MCP (Model Context Protocol), Multithreading, Concurrency

PROFESSIONAL EXPERIENCE:

Software Development Engineer (SDE) — Tata Consultancy Services | Jan 2025 – Present

- Designed and developed 13+ REST APIs powering a high-traffic production system serving 10,000+ requests/day, reducing response latency by 33% through performance optimization.
- Optimized SQL queries and data processing pipelines, improving application performance by 29% and ensuring reliability at scale.
- Enhanced retrieval-augmented generation (RAG) quality for a production RBI chatbot — tuning vector search relevance and building automated evaluation frameworks that improved answer accuracy by 19%.
- Collaborated with peer engineers through code reviews, upholding best practices for code quality, testability, and maintainability across the codebase.
- Implemented authentication and authorization using secure access-control frameworks aligned with enterprise security standards.

EDUCATION:

Bachelors of Technology in Artificial Intelligence and Data Science | Pune University | July 2020 to June 2024 | GPA: 8.88/10

CERTIFICATIONS:

- Microsoft Certified: Azure Data Scientist Associate [\[Link\]](#) 2026.
- AWS Certified Developer – Associate [\[Link\]](#) 2026.

PROJECTS:

Distributed Rate Limiter & API Gateway

Tech Stack: Go, gRPC/REST, Redis, PostgreSQL, Docker, Azure Kubernetes Service (AKS), Prometheus/Grafana, React

- Engineered a distributed rate-limiting gateway in Go using token-bucket and sliding-window algorithms, sharded via consistent hashing with Raft-based leader election for automatic failover.
- Exposed gRPC/REST APIs backed by Redis & PostgreSQL; load-tested to 45,000 req/sec at sub-8ms p99 latency using k6.
- Deployed on Azure Kubernetes Service with Prometheus/Grafana observability and a React dashboard for live traffic and quota monitoring.

Agentic AI Orchestration Platform

Tech Stack: Python, FastAPI, LLM tool-calling, Vector DB (Pinecone/Weaviate), PostgreSQL, Docker, React, WebSockets

- Designed a multi-agent orchestration framework with planner, retriever, executor, and critic agents coordinating through tool-calling and shared memory.
- Built a sandboxed, self-correcting execution loop, benchmarked with a Ragas-style evaluation harness achieving 87% task success across 150 test cases.
- Streamed live agent reasoning traces and tool calls to a React dashboard over WebSockets, with full audit logging in PostgreSQL for governance and replay.

High-Performance LLM Inference Server

Tech Stack: Rust, Python (PyO3 bindings), gRPC, Docker, React

- Built a custom LLM inference server in Rust implementing continuous batching and KV-cache management to maximize concurrent request throughput.
- Added dynamic batching and quantized (INT8/FP16) model loading, improving throughput by 230% and cutting p99 latency by 42% versus naive single-request serving.
- Exposed a gRPC interface with a React dashboard visualizing real-time token throughput, batch utilization, and memory footprint.